What is the leading cause of cancer-related death in the U.S. and worldwide?

Answer: Lung Cancer

Lung Cancer Survival

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Research Question(s)

Can unsupervised, clinically informed dimension reduction uncover meaningful lung cancer phenotypes?

Do these phenotypes meaningfully relate to survival outcomes?

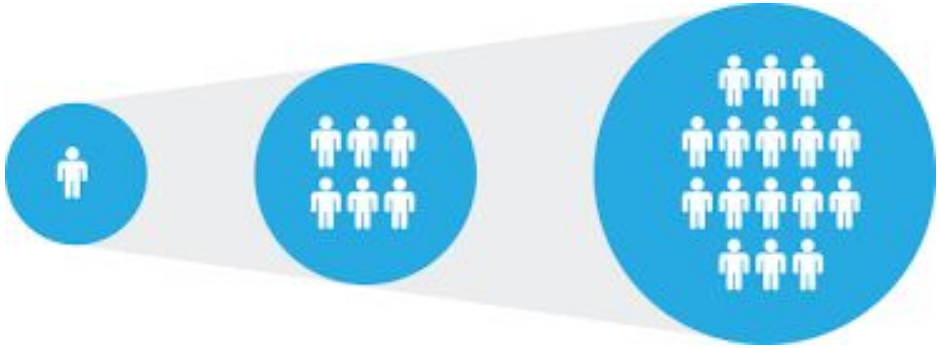
Data Source

- National Cancer Database (NCDB)
  - 70% of U.S. cancer cases
  - Multi-institution, real-world data
- Key Variables
  - Demographics
    - Age, sex, race, insurance
  - Tumor Characteristics
    - Histology, tumor size, stage, metastases
  - Health Status
    - Comorbidities, smoking
  - Treatment
    - Surgery, chemotherapy, immunotherapy
  - Outomes
    - Overall survival, vital status
  - Facility Characteristics
    - Type, location

NATIONAL CANCER DATABASE

Citation: American College of Surgeons & American Cancer Society (2024). National Cancer Database (NCDB). Participant User File. Retrieved March, 2026

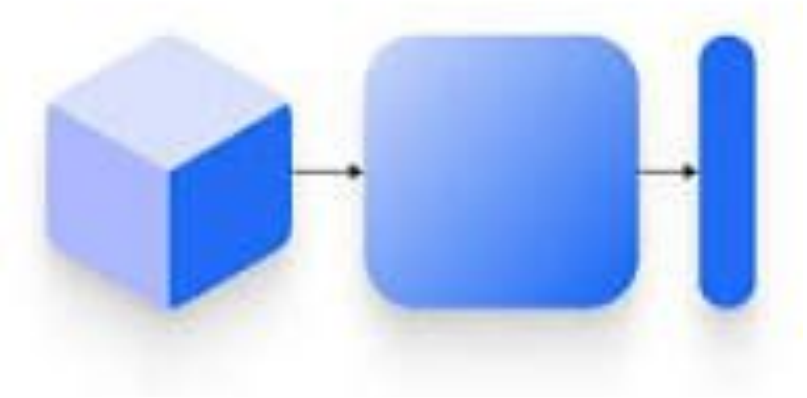
Challenges



- Scale
  - 2.5 million observations, +400 features
  - Subsetted to 500,000 observations, +30 features
    - ***That's still a lot!***
- Mixed variable types
- Clustering is not straightforward/feasible for the entire dataset

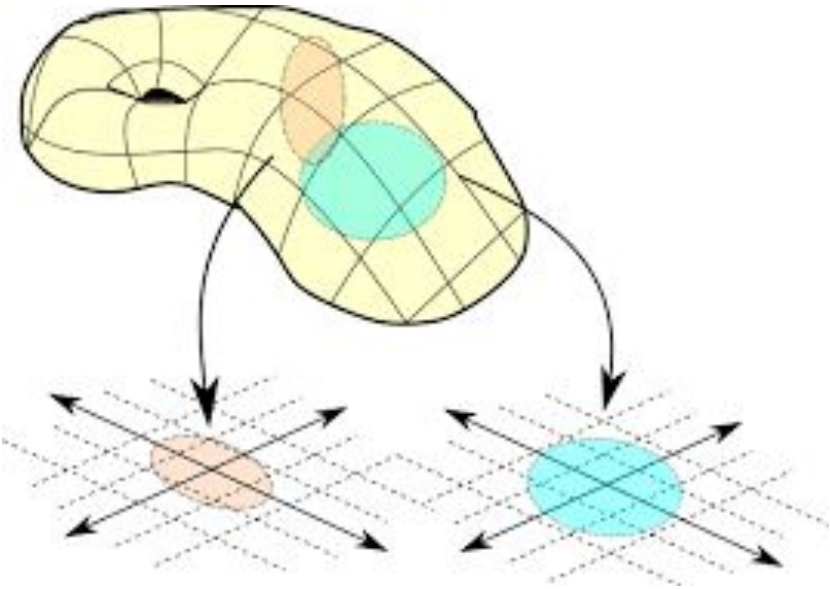
Dimensionality Reduction

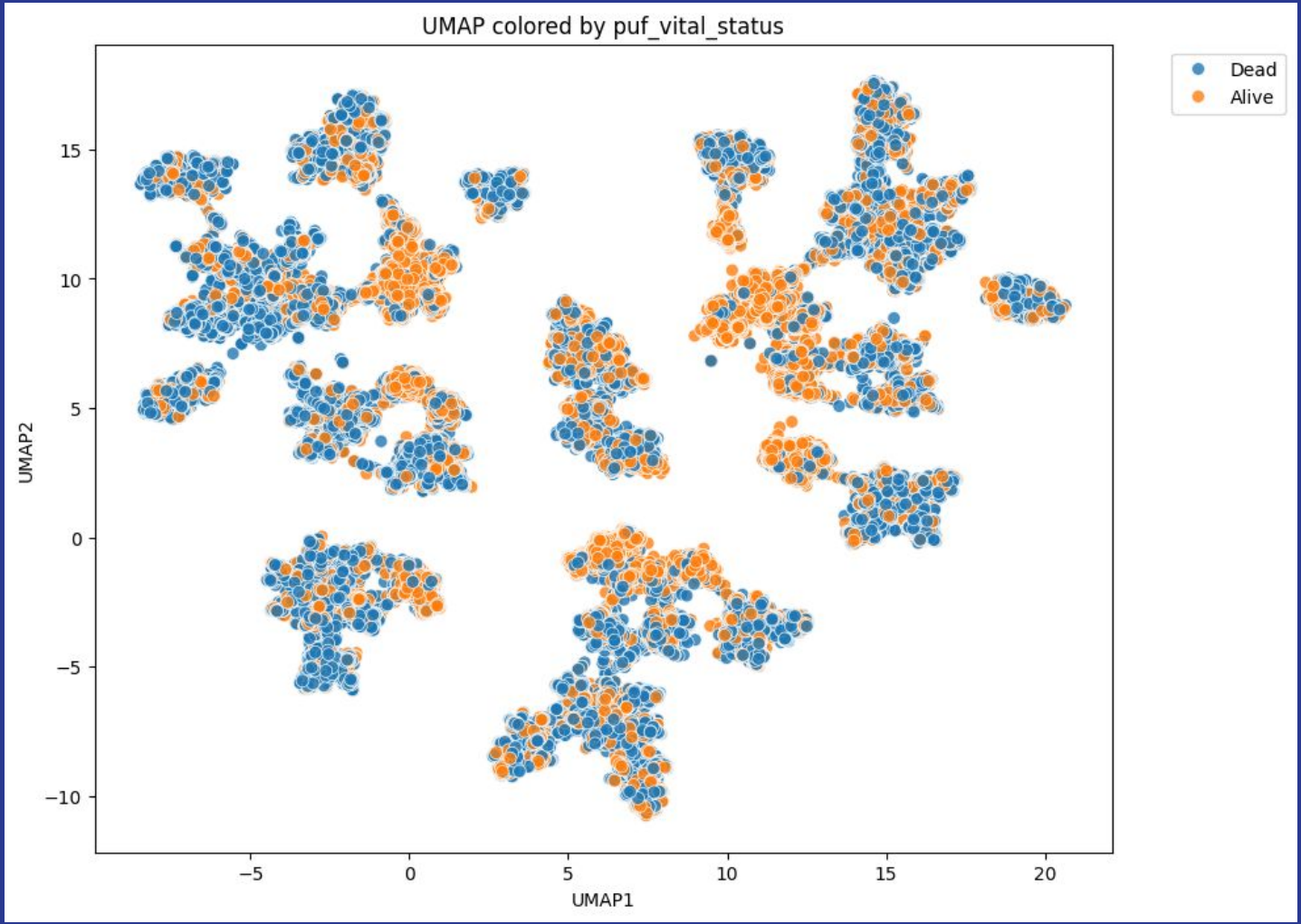
- **Clinically constrained, block-wise design**
  - Demographics & social context
  - Baseline health and risk
  - Tumor biology
  - Initial treatment strategy
- Separate reduction within each block
  - PCA, MCA (Multiple Correspondence Analysis), and FAMD (Factor Analysis of Mixed Data)
- Preserves clinical interpretability while reducing noise

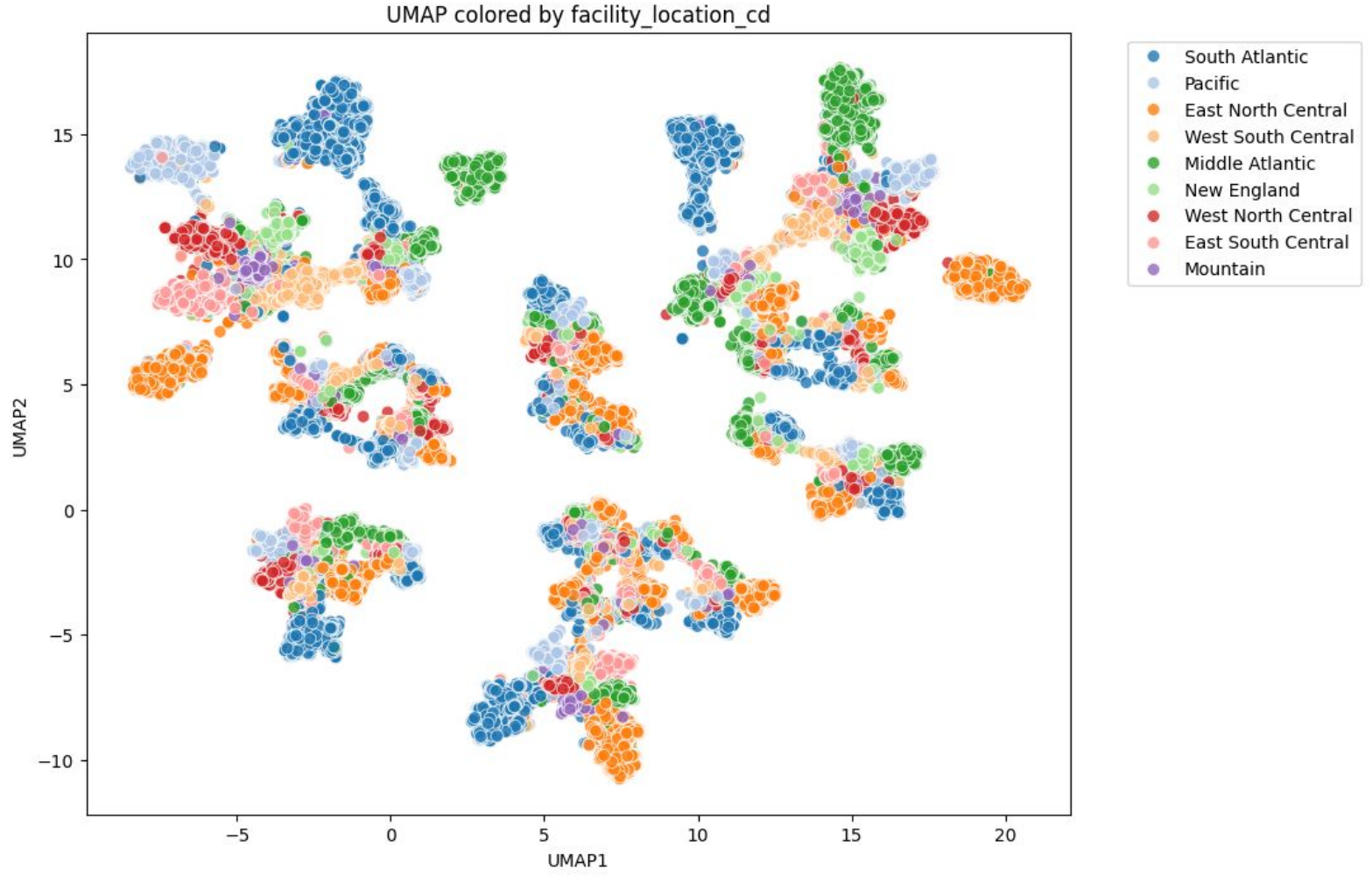


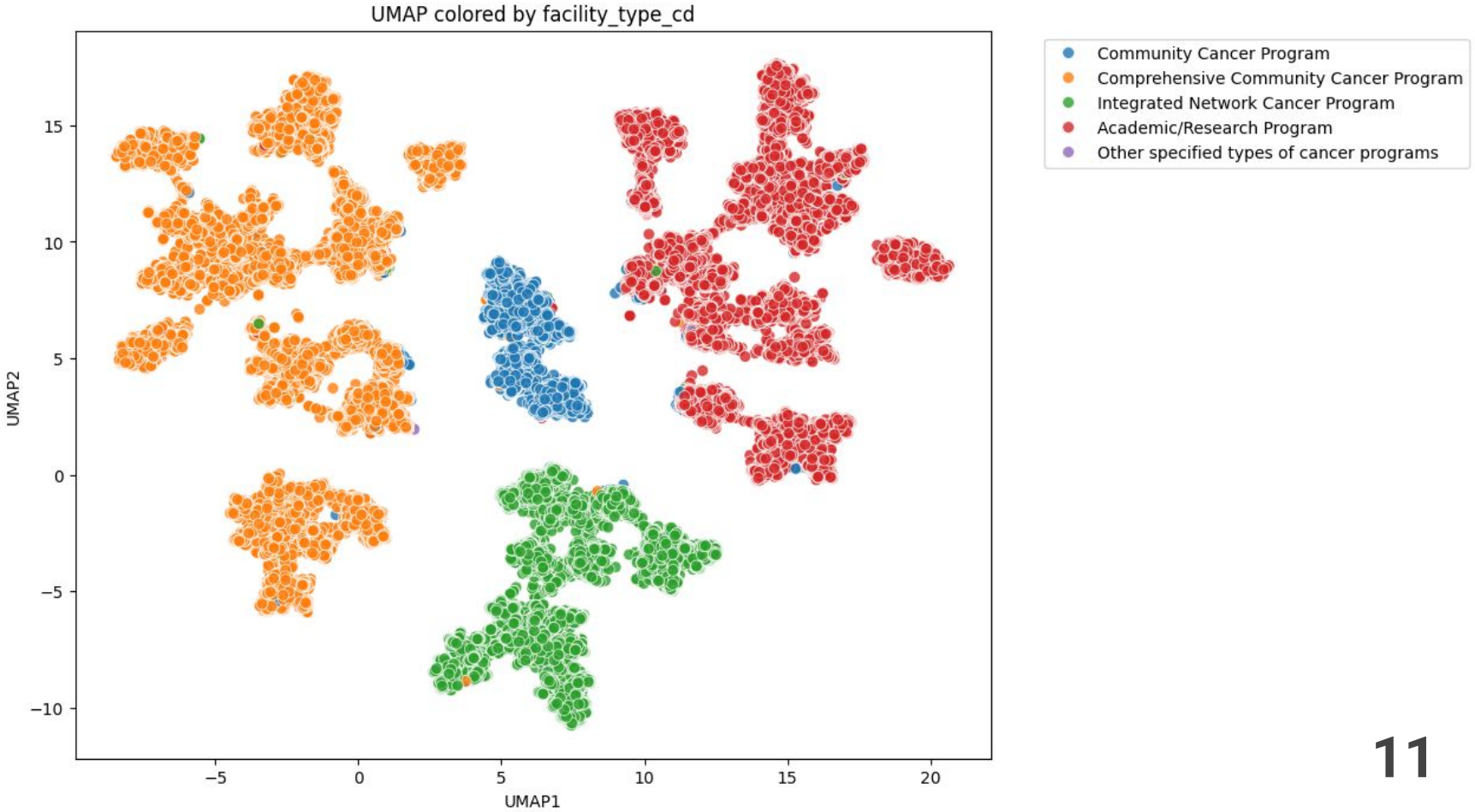
UMAP

- Uniform Manifold Approximation and Projection
- Applied to reduced feature representation
- Low-dimensional visualization of patient similarity
  - Each point represents a patient
  - Spatial proximity reflects similarity across clinical features



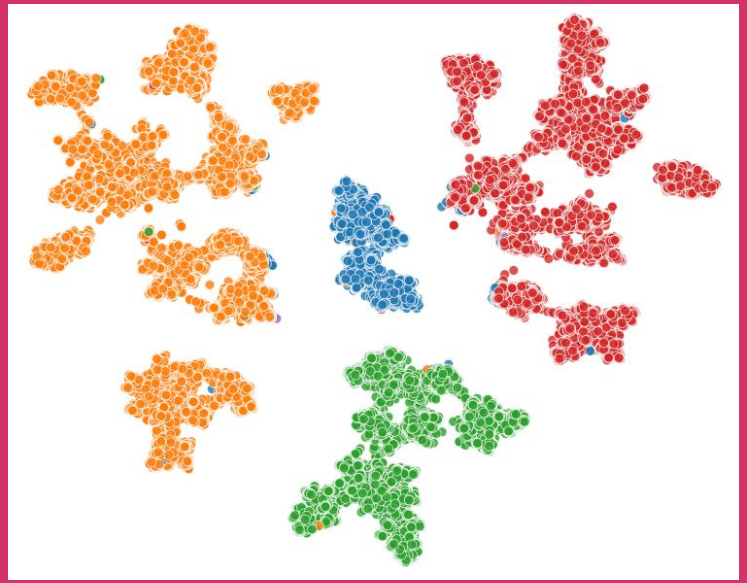

UMAP colored by puf_vital_status
Dead
Alive
UMAP2
UMAP1
15
10
5
0
−5
−10
−5
0
5
10
15
20

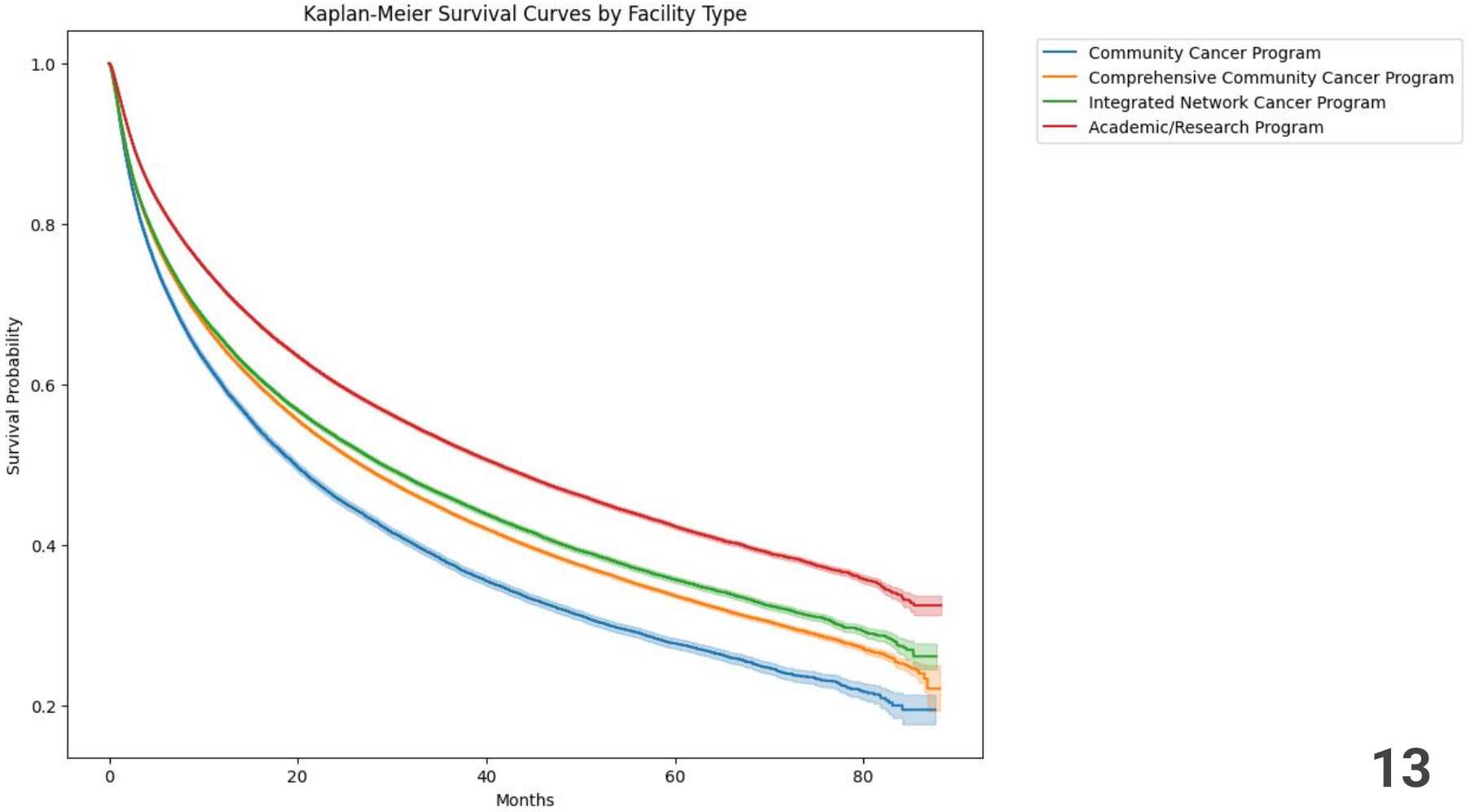

UMAP colored by facility_location_cd
South Atlantic
Pacific
East North Central
West South Central
Middle Atlantic
New England
West North Central
East South Central
Mountain
UMAP2
UMAP1
15
10
5
0
−5
−10
−5
0
5
10
15
20


UMAP colored by facility_type_cd
Community Cancer Program
Comprehensive Community Cancer Program
Integrated Network Cancer Program
Academic/Research Program
Other specified types of cancer programs
UMAP2
UMAP1
15
10
5
0
−5
−10
−5
0
5
10
15
20

11

System-level context appears to align with patient phenotypes




Kaplan-Meier Survival Curves by Facility Type
Survival Probability
1.0
0.8
0.6
0.4
0.2
0
20
40
60
80
Months
Community Cancer Program
Comprehensive Community Cancer Program
Integrated Network Cancer Program
Academic/Research Program

Next Steps

- Formal clustering on reduced feature space
- Compare survival between clusters through Cox Proportional Hazards modeling
- Identify phenotypes with greatest risk

$$\begin{aligned}\lambda(t|X_i) &= \lambda_0(t)\exp(\beta_1 X_{i1} + \cdots + \beta_p X_{ip}) \\ &= \lambda_0(t)\exp(X_i \cdot \beta)\end{aligned}$$

Questions?